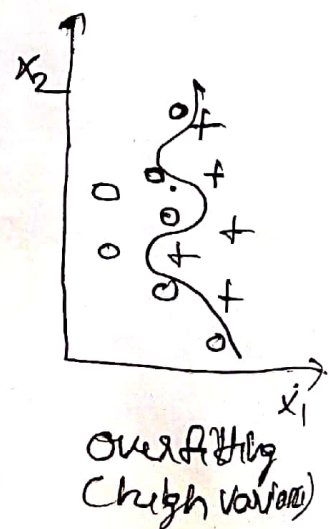
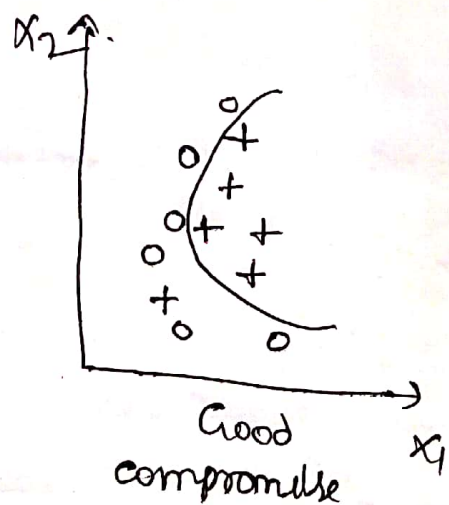
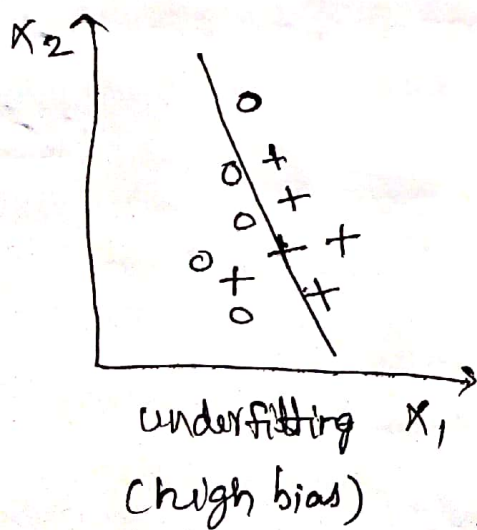


## over fit, under fit

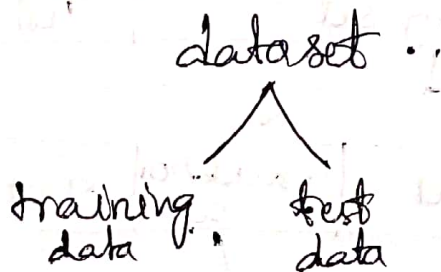
- Overfitting is a problem in machine learning, where a model performs well on training data but does not generalize well to unseen data (test data).
- If a model suffers from overfitting, means the model has high variance, caused by too many parameters.
- If a model suffer from underfitting (high bias), means model is not complex enough to capture the pattern in the training data.
  - suffers from low variance on unseen data.



- Regularization is very useful method to handle collinearity (high correlation among features), filter out noise from data, and eventually prevent overfitting.
- high bias  $\uparrow$  under fit  $\downarrow$  → model is too simple
- high variance  $\uparrow$  over fit  $\uparrow$  → model is too complex.

## holdout method

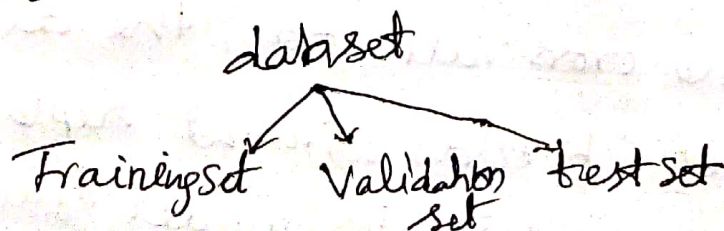
→ popular approach for estimating the generalization performance of machine learning model as hold out cross-validation.



Note:- In machine learning applications, data is tuned and compared different parameter settings to improve the performance for making prediction on unseen data. This process is known as model selection, where the term model selection refers to a given classification problem for which to select the optimum values of tuning parameters. (hyper)

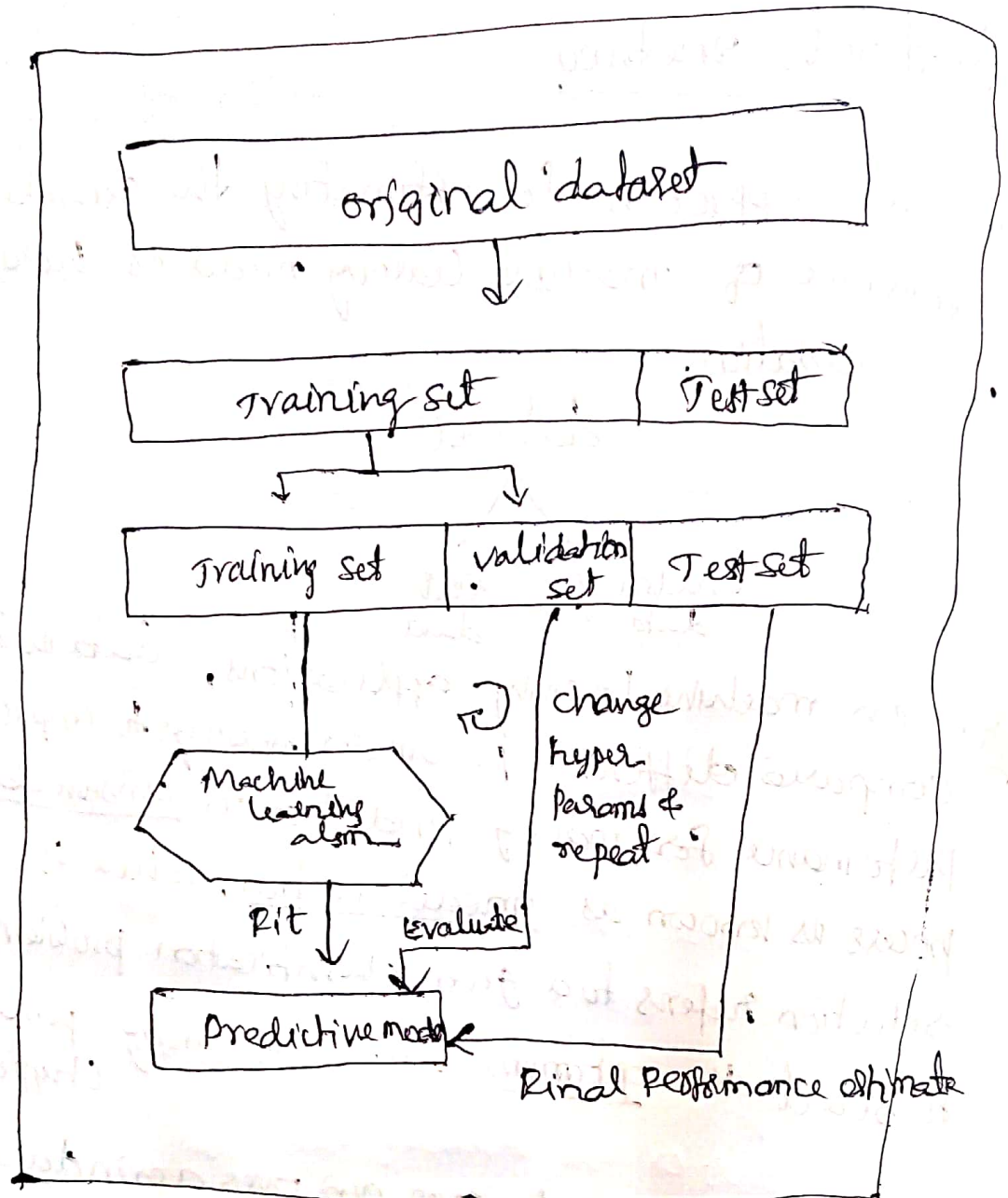
→ Using test data set over and over again during model selection, it will become part of training data and model will be more likely to overfit.

✗✗✗  
→ A better way of using holdout method for model selection is to separate data into three parts



→ training set is used to fit the models and performance on validation set ~~and~~ as then used for model selection. test set is used for generalizing performance of model.





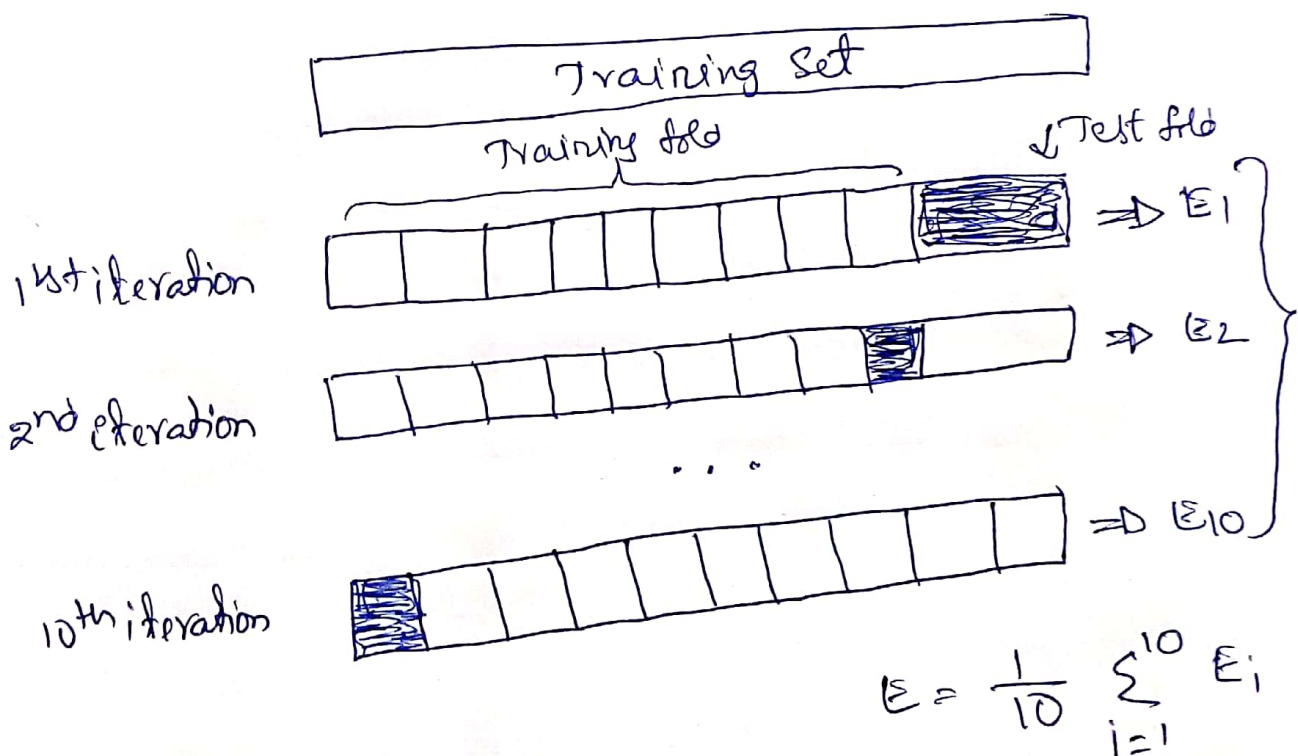
Disadvantage of hold out method is that performance estimate may be very sensitive to how the data is partitioned.

### K-Fold Validation:-

- In k-fold cross validation, the dataset is randomly split into k folds without replacement, where k-1 folds are used for model training, and one fold is used for model performance evaluation.
- This process is repeated k times to obtain k models and performance estimates.

→ Then calculate the average performance of model

→ Since  $k$ -fold cross validation is a resampling technique without replacement, the advantage of this approach is that each sample point will be used for training and validation exactly once, which yields a lower variance estimate of the model performance than the hold out method.



→ Research shows that 10 fold-cross validation offers best trade off between bias and variance.

\* If working with large datasets choose a smaller value of  $k$ , ( $k=5$ ) to obtain an accurate estimate of the average performance of the model while reducing the computational cost.



# \*\*\* Overfitting, underfitting

## Bias and Variance

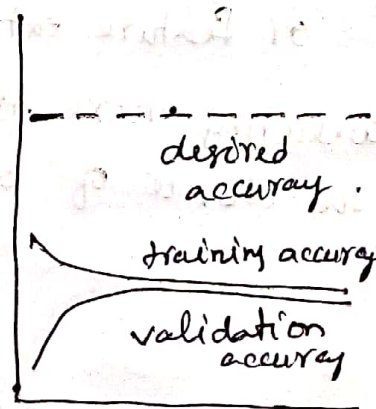
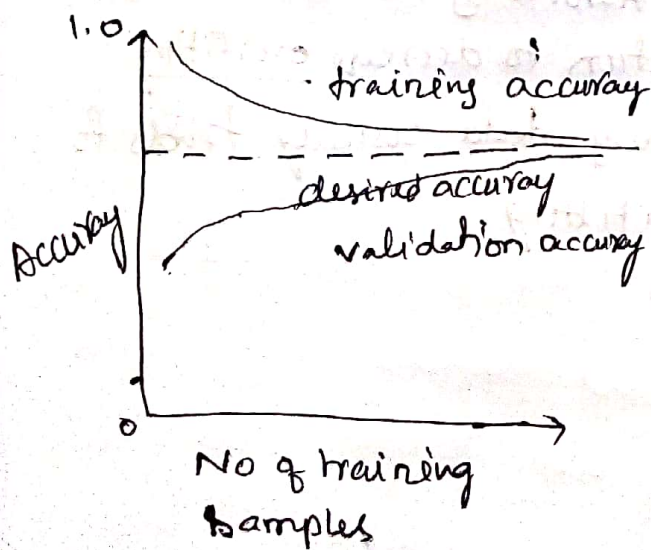
→ If a model is too complex for a given training dataset there are too many degrees of freedom or parameters in the model.

→ the model tends to overfitting the training data and does not generalize well to unseen data

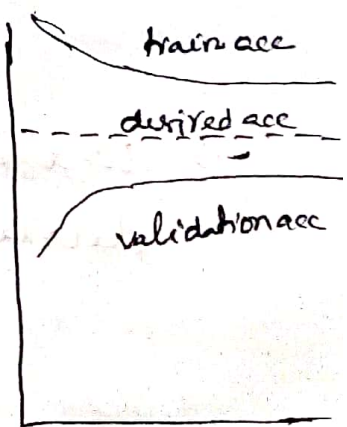
→ need more training data to reduce the degree of overfitting.

\* In practice it can often be expensive or not possible to collect more data.

→ By plotting the model training and validation accuracies as functions of the training set size, it can be detected whether the model suffers from high variance or high bias



High bias



High Variance

Good bias variance trade off

## High bias :-

→ The model has both low training and cross validation accuracy, which indicates that it underfits the training data.

Sol:

→ Increase the number of parameters of the model  
(add additional features)  
or by decreasing the degree of regularization  
(SVM or logistic regression)

## High Variance :-

→ Indicates the large gap between the training and cross validation accuracy.

Sol :-

→ collect more training data, reduce the complexity of the model or increase the regularization parameters.

(decrease the no of features by Feature Selection)  
or Feature extraction to decrease overfitting

→ While collecting more training data usually tends to decrease the chance of overfitting.